Notes: Ossa (JPE, 2011)

A "New Trade" Theory of GATT/WTO Negotiations

Contents

1 Big picture

- **Question.** Why do countries need to negotiate over tariffs, and what exactly do the GATT/WTO principles of **reciprocity** and **nondiscrimination** do?
- **Setting.** The paper replaces the standard neoclassical optimal-tariff framework with a **Krugman (1980) new-trade model** featuring monopolistic competition, increasing returns, and endogenous relocation of manufacturing firms across countries.
- **Core challenge.** In the standard terms-of-trade view, countries negotiate because they otherwise manipulate world prices. Ossa argues that this is not a very convincing description of actual trade-policy rhetoric, and it is also hard to calibrate quantitatively.
- **Main theoretical result.** In a new-trade environment, unilateral import tariffs create a **production relocation externality**: by raising tariffs, a country can attract manufacturing firms and jobs away from other countries. Reciprocity neutralizes this externality by fixing manufacturing trade balances, and nondiscrimination makes this internalization work in a multilateral setting.
- **Main institutional interpretation. Reciprocity** is what reverses incentives for protection and guides countries away from the inefficient noncooperative equilibrium. **Nondiscrimination** does not add efficiency directly; instead, it **multilateralizes** negotiations so third-country externalities are internalized as well.
- **Main quantitative result.** Calibrated to 2004 bilateral trade and tariff data for Brazil, China, the European Union, India, Japan, the United States, and the rest of the world, the model implies Nash tariffs of about **10–13%** when the elasticity of substitution is high and about **26–29%** when it is lower. Welfare losses from moving from factual tariffs to Nash tariffs are **moderate**, and usually amount to only a fraction of the welfare losses from autarky.
- **Why this paper matters.** It provides a clean alternative microfoundation for trade agreements, one that emphasizes **jobs, firm location, and industrial relocation** rather than only terms-of-trade manipulation.

2 Data and measurement (key objects)

2.1 Observed country-level trade and tariff panel

- **Unit of observation.** A country-pair or trade-bloc pair (i, j) in a static cross section centered on **2004**.
- **Countries in the calibration.** The paper focuses on the major players in recent GATT/WTO negotiations: **Brazil, China, the European Union, India, Japan, and the United States**, plus a seventh block equal to the **rest of the world (ROW)**.
- **Observed bilateral trade flows.** The key input is the matrix of bilateral manufacturing trade flows T_{ij}, evaluated at world prices. Internal trade flows are also constructed, following Dekle, Eaton, and Kortum.

- **Observed tariffs.** The paper uses importer-specific manufacturing tariffs in 2004, based on applied protection rates from Boumellassa, Laborde, and Mitaritonna. Because of nondiscrimination, these tariffs vary by importer only in the calibration.

- **Country size and expenditure objects.** Labor endowments or consumer masses L_j, trade surpluses TB_j, total expenditure X_j, and tariff revenue R_j enter the multi-country system.

2.2 Parameters disciplined from outside the model

- **Manufacturing expenditure share.** The calibration uses

$$\mu = 0.188.$$

- **Elasticity of substitution.** The paper considers two benchmark values,

$$\sigma = 9.28 \quad \text{and} \quad \sigma = 4.60,$$

  in order to show how quantitative implications vary with demand elasticity.

- **Technology and trade-cost heterogeneity.** In the quantitative model, marginal costs c_i, fixed costs f_i, and bilateral transport costs v_{ij} are allowed to differ across countries.

2.3 Key measured objects

- **Manufacturing price index** G_j: summarizes the consumer price index for differentiated manufactures in country j.

- **Number of manufacturing firms** n_j: the location of firms across countries is central because tariffs shift profitability and therefore entry and exit.

- **Manufacturing trade balance** TB_j^M: this is the crucial object in the reciprocity argument, because holding it fixed prevents tariff-induced relocation of firms.

- **Welfare.** Welfare is measured through indirect utility, which depends on expenditure and the manufacturing price index.

2.4 Unobservables and timing

- **Latent equilibrium objects.** For any tariff vector, the equilibrium numbers of firms n_j, manufacturing price indices G_j, and expenditure levels X_j are endogenous and solved from the model.

- **Static timing.** The model is static in the sense that it compares alternative tariff equilibria rather than solving a dynamic path. The key endogenous margin is **entry and exit of firms across countries**.

- **Counterfactual interpretation.** The quantitative section interprets factual 2004 trade patterns as one equilibrium and compares them to counterfactual Nash tariffs, autarky, and additional reciprocal liberalization.

2.5 Empirical applications

- **Qualitative application.** Reinterpret reciprocity and nondiscrimination as rules that internalize production relocation rather than terms-of-trade externalities.
- **Quantitative application.** Compute noncooperative tariffs and welfare losses for the main WTO players.
- **Policy application.** Evaluate how much more mutually beneficial liberalization remains after decades of GATT/WTO tariff cuts.

3 Models and empirical specifications (all equations + interpretation)

3.1 Basic two-country model

There are two countries, indexed by $j \in \{1, 2\}$. Consumers in country j consume a homogeneous nonmanufacturing good Y_j and a continuum of differentiated manufacturing varieties produced in each country.

Preferences are

$$U_j = \left[\sum_{i=1}^{2} \int_0^{n_i} m_{ij}(u)^{(\sigma-1)/\sigma} \, du \right]^{\mu\sigma/(\sigma-1)} Y_j^{1-\mu},$$

where $m_{ij}(u)$ is consumption in country j of a manufacturing variety from country i, n_i is the measure of manufacturing varieties produced in country i, $\sigma > 1$ is the elasticity of substitution across varieties, and μ is the expenditure share on manufactures.

Production technologies are

$$l_j^M = f + c q_j^M,$$
$$l_j^Y = q_j^Y,$$

where manufacturing has a fixed cost f and marginal labor requirement c, while the outside good is linear in labor.

Manufacturing is monopolistically competitive, the outside sector is perfectly competitive, and tariffs are embedded in iceberg trade costs. To deliver one unit from i to j, firms must ship

$$v(1 + t_{ij})$$

units, where $v > 1$ is a transport-cost term and $t_{ij} \geq 0$ is the import tariff imposed by country j on country i.

- **Interpretation.** This is a Krugman-style setting where tariff changes affect not only consumer prices but also the **location of production** because firm profits induce entry and exit.
- **Why the outside good matters.** Because the nonmanufacturing good is freely traded and always produced, wages are pinned down at one, so the model isolates **production relocation effects** rather than terms-of-trade effects through wages.

3.2 Equilibrium for given tariffs

With the outside good as numeraire, wages equal one in both countries. Free entry and monopolistic pricing imply

$$q = \frac{f(\sigma-1)}{c}, \qquad p = \frac{\sigma c}{\sigma-1}.$$

Manufacturing market clearing in the two-country case is

$$q = p^{-\sigma} G_1^{\sigma-1} \mu L_1 + p^{-\sigma}(vt_{12})^{1-\sigma} G_2^{\sigma-1} \mu L_2,$$

$$q = p^{-\sigma}(vt_{21})^{1-\sigma} G_1^{\sigma-1} \mu L_1 + p^{-\sigma} G_2^{\sigma-1} \mu L_2.$$

The manufacturing price indices are

$$G_1 = \left[n_1 p^{1-\sigma} + n_2 (pvt_{21})^{1-\sigma}\right]^{1/(1-\sigma)},$$

$$G_2 = \left[n_1 (pvt_{12})^{1-\sigma} + n_2 p^{1-\sigma}\right]^{1/(1-\sigma)}.$$

Solving gives

$$G_1 = \left[\frac{qp}{\mu L_1} \cdot \frac{1-(vt_{12})^{1-\sigma}}{1-(vt_{21}vt_{12})^{1-\sigma}}\right]^{1/(\sigma-1)},$$

$$G_2 = \left[\frac{qp}{\mu L_2} \cdot \frac{1-(vt_{21})^{1-\sigma}}{1-(vt_{21}vt_{12})^{1-\sigma}}\right]^{1/(\sigma-1)}.$$

The corresponding numbers of firms are

$$n_1 = \frac{\mu}{qp}\left[\frac{L_1}{1-(vt_{12})^{1-\sigma}} - \frac{L_2 (vt_{21})^{1-\sigma}}{1-(vt_{21})^{1-\sigma}}\right],$$

$$n_2 = \frac{\mu}{qp}\left[\frac{L_2}{1-(vt_{21})^{1-\sigma}} - \frac{L_1 (vt_{12})^{1-\sigma}}{1-(vt_{12})^{1-\sigma}}\right].$$

Indirect utility is

$$V_j = \mu^{\mu}(1-\mu)^{1-\mu} L_j G_j^{-\mu}.$$

- **Interpretation.** Welfare depends on the manufacturing price index. Tariffs therefore matter because they reshape the location of firms and the share of varieties that consumers access without paying trade costs.
- **Special feature.** The **world number of firms** is fixed in the basic model, so tariffs reallocate firms across countries rather than changing the global measure of varieties.

3.3 Production relocation effect versus import price effect

A unilateral tariff increase by country j has two effects on its own manufacturing price index.

- **Import price effect.** Still-imported foreign goods become more expensive, which pushes the domestic price index up.

- **Production relocation effect.** Domestic consumers switch expenditure toward home-produced manufacturing goods. Domestic firms become more profitable, entry occurs at home, exit occurs abroad, and the domestic price index falls while the foreign price index rises.

In Ossa's basic model, the production relocation effect dominates the import price effect. Therefore a country's own tariff lowers its own price index and raises the foreign price index.

- **Interpretation.** This is the central externality in the paper. A government can improve domestic welfare by pulling firms and manufacturing jobs toward itself, even though this hurts foreigners.
- **Why this is distinct from terms of trade.** Ex-factory prices do not move with tariffs in the basic model, so the gain cannot be interpreted as a standard terms-of-trade effect.

3.4 Noncooperative tariffs

If governments choose tariffs noncooperatively to maximize domestic welfare, the paper shows:

- **Lemma 1.** The unique Nash equilibrium in the basic two-country model is maximum protection:

  $$(t_{21}, t_{12}) = (\bar{t}, \bar{t}),$$

  where $\bar{t}$ is the upper bound on tariffs.
- **Lemma 2.** The set of Pareto-efficient tariff pairs consists of tariff combinations where at least one country sets a zero tariff.
- **Proposition 1.** The noncooperative equilibrium is inefficient.
- **Interpretation.** Since each country ignores the damage its tariff does to foreign price indices by relocating firms, Nash tariffs are too high.
- **Economic meaning.** The paper generates a prisoner's-dilemma logic without relying on terms-of-trade manipulation.

3.5 Reciprocity in the two-country model

The paper formalizes reciprocity by requiring that manufacturing trade balances remain unchanged.

Definition (reciprocity). A tariff change is reciprocal if

$$d\mathrm{TB}_1^M = d\mathrm{TB}_2^M = 0.$$

The key accounting identity is

$$n_j = \frac{\mu L_j + \mathrm{TB}_j^M}{qp}.$$

Hence, if reciprocity keeps manufacturing trade balances fixed, it also keeps the number of firms in each country fixed.

- **Lemma 3.** Tariff changes leave the number of firms unchanged in both countries if and only if they are reciprocal.

- **Proposition 2.** Reciprocal trade liberalization monotonically increases welfare in both countries.
- **Interpretation.** Reciprocity shuts down the production relocation effect. Once firm counts are fixed, only the import-price effect remains, so lower tariffs unambiguously improve welfare.
- **Institutional implication.** Reciprocity not only moves countries away from the inefficient Nash equilibrium; it also removes incentives to backslide because retaliation restores the same manufacturing trade-balance discipline.

3.6 Three-country extension and nondiscrimination

To study nondiscrimination, Ossa extends the model to three countries where country 1 trades with countries 2 and 3, while countries 2 and 3 trade only with country 1. Country 1 can therefore discriminate across partners.

Definition (bilateral and multilateral reciprocity). A tariff change is bilaterally reciprocal between countries 1 and 2 if

$$d\mathrm{TB}_2^M = 0,$$

and bilaterally reciprocal between countries 1 and 3 if

$$d\mathrm{TB}_3^M = 0.$$

It is multilaterally reciprocal if

$$d\mathrm{TB}_1^M = d\mathrm{TB}_2^M = d\mathrm{TB}_3^M = 0.$$

The same accounting identity holds:

$$n_j = \frac{\mu L_j + \mathrm{TB}_j^M}{qp}.$$

- **Lemma 4.** Tariff changes leave firm counts unchanged in all countries if and only if they are **multilaterally** reciprocal. Bilaterally reciprocal liberalization between 1 and 2 leaves country 2's firm count unchanged but increases country 1's firms at the expense of country 3.
- **Proposition 3.** Multilaterally reciprocal trade liberalization monotonically increases welfare in all countries, whereas bilaterally reciprocal liberalization can hurt excluded third countries.
- **Interpretation.** Bilateral reciprocity solves only the direct externality between negotiating partners. It does not prevent tariff changes from affecting third-country profitability through price-index changes in the hub country.
- **Role of nondiscrimination.** Nondiscrimination forces tariff changes to apply to all partners symmetrically, which effectively **multilateralizes reciprocity**. That is Ossa's explanation for why nondiscrimination matters in the WTO architecture.
- **Important nuance.** Nondiscrimination does not itself create the efficiency gain. Reciprocity does. Nondiscrimination just ensures that reciprocity works **for everyone**, not only for the countries at the negotiating table.

3.7 Quantitative multi-country model

The calibrated model allows for J countries, asymmetric technologies, bilateral transport costs, tariff revenue, and exogenous aggregate trade imbalances.

For given tariffs, the equilibrium is characterized by

$$q_i = \sum_{j=1}^{J} p_i^{-\sigma} v_{ij}^{1-\sigma} t_{ij}^{-\sigma} G_j^{\sigma-1} \mu X_j, \qquad i = 1, \ldots, J,$$

$$G_j = \left[\sum_{i=1}^{J} n_i (p_i v_{ij} t_{ij})^{1-\sigma} \right]^{1/(1-\sigma)}, \qquad j = 1, \ldots, J,$$

$$X_j = L_j - TB_j + \sum_{i=1}^{J} t_{ij} n_i (p_i v_{ij})^{1-\sigma} t_{ij}^{-\sigma} G_j^{\sigma-1} \mu X_j, \qquad j = 1, \ldots, J.$$

The last equation says consumer expenditure equals labor income minus the trade surplus plus tariff revenue.

Using the Dekle-Eaton-Kortum change formulation, counterfactuals can be written as

$$1 = \sum_{j=1}^{J} a_{ij} \hat{t}_{ij}^{-\sigma} \hat{G}_j^{\sigma-1} \hat{X}_j, \qquad i = 1, \ldots, J,$$

$$\hat{G}_j = \left[\sum_{i=1}^{J} b_{ij} \hat{n}_i \hat{t}_{ij}^{1-\sigma} \right]^{1/(1-\sigma)}, \qquad j = 1, \ldots, J,$$

$$\hat{X}_j = g_j + \sum_{i=1}^{J} d_{ij} t'_{ij} \hat{n}_i \hat{t}_{ij}^{-\sigma} \hat{G}_j^{\sigma-1} \hat{X}_j, \qquad j = 1, \ldots, J,$$

where $a_{ij}, b_{ij}, g_j, d_{ij}$ are functions of observed trade flows and tariffs only.

Counterfactual welfare changes are then

$$\hat{V}_j = \hat{X}_j \hat{G}_j^{-\mu}.$$

- **Interpretation.** This is the paper's main quantitative contribution: once you observe bilateral trade and tariffs, you can solve directly for equilibrium changes in firm location, price indices, expenditure, and welfare under alternative tariff vectors.
- **Why this matters.** Compared with conventional optimal-tariff models, the calibration is unusually transparent because it leans directly on bilateral trade data.

3.8 Calibration and counterfactual mapping

- **Trade blocs.** The baseline calibration includes Brazil, China, the European Union, India, Japan, the United States, and the rest of the world.
- **Trade data.** Bilateral manufacturing trade flows come from the aggregation method of Dekle, Eaton, and Kortum.

- **Tariff data.** Factual 2004 tariffs are simple averages of applied manufacturing protection rates.
- **Elasticity cases.** The paper reports results for $\sigma = 9.28$ and $\sigma = 4.60$.
- **Computation of Nash tariffs.** Ossa illustrates an iterative best-response algorithm in a simple United States-versus-rest-of-world case and then applies the same logic in the full seven-bloc environment.

4 Identification and instruments (what makes the estimates credible)

4.1 What fails in the standard approach

- **Problem with the standard theory.** The standard Bagwell-Staiger logic says countries negotiate to prevent terms-of-trade manipulation. Ossa argues that this is empirically unsatisfying because trade-policy debates are usually framed in terms of **production, jobs, and industrial location**, not export-price manipulation.
- **Problem with standard calibration.** Conventional neoclassical trade-agreement models are hard to calibrate convincingly, so their quantitative implications for Nash tariffs and welfare losses are not very disciplined.

4.2 What identifies the new mechanism in the theory

- **Key theoretical identification.** The model strips out wage-based terms-of-trade movements by using a freely traded homogeneous outside good with unit labor productivity. This pins wages at one and isolates **production relocation** as the operative externality.
- **Trade-balance channel.** Reciprocity is shown to matter because, through

$$n_j = \frac{\mu L_j + \mathrm{TB}_j^M}{qp},$$

  holding the manufacturing trade balance fixed also holds the domestic firm mass fixed.
- **Multilateral logic.** In the three-country setup, the paper identifies third-country spillovers through changes in the hub country's price index. This is what gives nondiscrimination a role.

4.3 What pins down the quantitative model

- **Observed bilateral trade flows** pin down the factual equilibrium shares that enter the counterfactual change system.
- **Observed tariffs** discipline the starting protection structure.
- **External parameter values** for μ and σ discipline demand curvature and the magnitude of firm-relocation responses.
- **Exogenous trade balances** allow the model to match the fact that countries need not be balanced in the data.

4.4 What is assumed rather than identified

- **Import tariffs only.** The paper abstracts from export taxes and export subsidies. That is deliberate, because real-world GATT/WTO practice is heavily centered on import tariffs.
- **No intermediate goods in the baseline.** This keeps the model tractable and avoids the cumulative-causation forces that would otherwise generate multiple equilibria in a Krugman-Venables style setting.
- **No direct wage adjustment margin in the basic model.** This helps isolate production relocation, but it also means the note should be read as highlighting a mechanism rather than offering a fully general trade model.

4.5 Relation to the literature

- **Against the Johnson-Bagwell-Staiger tradition.** The paper keeps the game-theoretic logic of trade agreements but replaces the source of the externality.
- **Related to Krugman-Venables new trade.** The externality comes from increasing returns and endogenous firm location.
- **Related quantitative strategy.** The calibration builds directly on Dekle, Eaton, and Kortum's exact-hat algebra style counterfactual method.

5 Tables and figures

5.1 Table 1: Aggregated trade matrix

Read:

- The quantitative exercise is grounded in a seven-bloc bilateral manufacturing trade matrix for 2004.
- Internal trade is very large for the biggest blocs, especially the European Union and the United States. This matters because firm relocation is easier to trigger when domestic markets are large.
- The table makes clear that the exercise is not a stylized symmetric calibration; it is built on strongly asymmetric trade relationships.

TABLE 1
AGGREGATED TRADE MATRIX FROM DEKLE ET AL. (2007)

	ROW	European Union	Brazil	China	India	Japan	United States
ROW	3,907.4	551.6	15.1	434.4	20.4	91.2	550.8
European Union	656.9	6,372.9	14.3	83.6	16.9	48.3	235.9
Brazil	24.1	9.3	314.6	2.1	.3	1.0	16.4
China	349.7	161.6	3.9	801.7	7.0	82.4	212.2
India	18.6	17.3	.4	6.2	387.0	1.4	14.6
Japan	191.8	96.1	2.9	123.1	2.9	3,074.1	128.4
United States	390.4	177.4	10.7	45.6	5.5	44.0	5,201.3

NOTE.—Entry (i, j) is factual exports from i to j in US$ billions in 2004.

TABLE 2
AGGREGATED TARIFF MATRIX FROM BOUMELLASSA ET AL. (2009)

	ROW	European Union	Brazil	China	India	Japan	United States
ROW	.0	2.5	12.7	4.2	14.8	1.3	2.2
European Union	7.0	.0	12.7	4.2	14.8	1.3	2.2
Brazil	7.0	2.5	.0	4.2	14.8	1.3	2.2
China	7.0	2.5	12.7	.0	14.8	1.3	2.2
India	7.0	2.5	12.7	4.2	.0	1.3	2.2
Japan	7.0	2.5	12.7	4.2	14.8	.0	2.2
United States	7.0	2.5	12.7	4.2	14.8	1.3	.0

NOTE.—Entry (i, j) is the factual tariff imposed by j against imports from i in percent in 2004.

5.2 Table 2: Aggregated tariff matrix

Read:

- Factual 2004 tariffs differ greatly across importers: the United States is very lightly protected at about **1.3%**, the European Union is at about **2.5%**, India at about **4.2%**, the European Union's partners face **7.0%** into the EU column, China is around **12.7%**, and Japan is the most protected at about **14.8%**.
- Because tariffs vary by importer only, the table is consistent with nondiscrimination in the calibration.
- These factual tariffs are already far below the model's predicted Nash tariffs, which is the key quantitative sense in which GATT/WTO negotiations have moved the world away from noncooperative protection.

5.3 Figure 1: Optimal tariff with tariff revenue

Read:

- In the simple United States-versus-rest-of-world example, welfare is hump shaped in the tariff once tariff revenue is included.
- This is important because it shows why the quantitative model no longer predicts literal maximum tariffs as Nash outcomes: tariff revenue eventually falls enough that further protection becomes counterproductive.
- So the interior optimum in the quantitative model is generated by a trade-off between more domestic firm relocation and lower tariff revenue at very high tariffs.

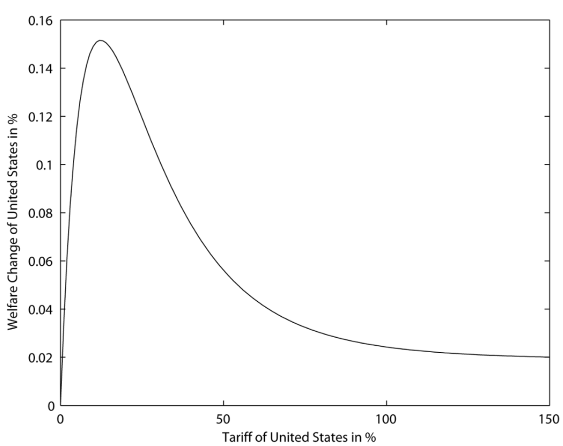

Fig. 1.—Optimal tariff with tariff revenue

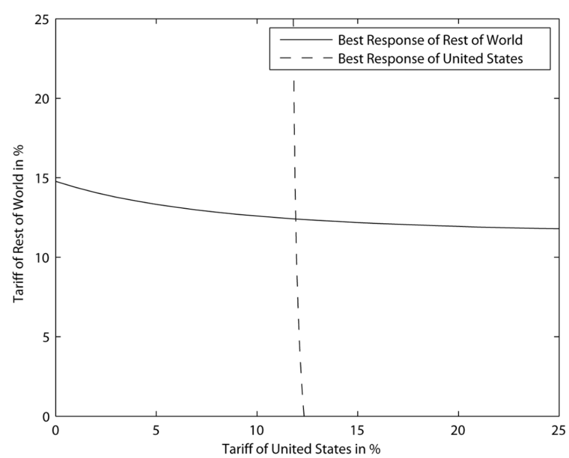

Fig. 2.—Reaction functions and Nash equilibrium

5.4 Figure 2: Reaction functions and Nash equilibrium

Read:

- The figure shows that Nash tariffs can be found with a simple iterative best-response algorithm.

- Intuitively, this matters because the model remains computationally manageable even in the multi-country case.
- It also visually reinforces the paper's message that strategic tariff setting is a genuine equilibrium object even in a new-trade environment.

5.5 Table 3: Nash tariffs with $\sigma = 9.28$

Read:

- When demand is relatively elastic, predicted Nash tariffs lie in a fairly tight band between about **10.3% and 13.0%**.
- The cross-country variation is modest, which says that the basic production-relocation motive is quantitatively similar across the major blocs in this calibration.
- The striking message is not the fine differences across countries, but that the model produces Nash tariffs in the **same order of magnitude** as historically observed tariff wars.

TABLE 3
NASH TARIFFS WITH $\sigma = 9.28$

	ROW	European Union	Brazil	China	India	Japan	United States
ROW	.0	13.0	12.1	11.6	12.0	12.4	12.4
European Union	11.0	.0	12.1	11.9	12.0	12.1	12.0
Brazil	11.6	12.5	.0	12.0	12.1	12.2	12.3
China	10.3	11.5	12.0	.0	12.0	12.5	10.9
India	12.2	12.5	12.1	12.0	.0	12.1	12.1
Japan	11.3	12.0	12.1	11.6	12.1	.0	11.9
Unitd States	11.4	12.1	12.1	12.0	12.1	12.1	.0

NOTE.—Entry (i, j) is the counterfactual Nash tariff imposed by j against imports from i in percent.

TABLE 4
NASH TARIFFS WITH $\sigma = 4.60$

	ROW	European Union	Brazil	China	India	Japan	United States
ROW	.0	28.7	27.8	26.2	27.7	28.3	27.8
European Union	26.3	.0	27.7	27.6	27.8	27.8	27.7
Brazil	26.7	28.2	.0	27.6	27.8	27.9	27.8
China	28.4	26.9	27.7	.0	27.7	28.9	25.6
India	27.5	28.0	27.8	27.5	.0	27.8	27.6
Japan	26.6	27.5	27.8	26.9	27.8	.0	27.5
United States	27.0	27.8	27.8	27.8	27.8	27.9	.0

NOTE.—Entry (i, j) is the counterfactual Nash tariff imposed by j against imports from i in percent.

5.6 Table 4: Nash tariffs with $\sigma = 4.60$

Read:

- Lower elasticity of substitution implies much higher Nash tariffs, around **25.6% to 28.9%**.
- This comparative-static is intuitive: when varieties are less substitutable, attracting firms and domestic production is more valuable, so protection incentives are stronger.
- The table is a reminder that quantitative conclusions depend importantly on the trade elasticity.

5.7 Table 5: Welfare losses from moving to Nash tariffs

Read:

- Moving from factual 2004 tariffs to Nash tariffs produces **moderate** welfare losses: between about **-0.03% and -0.36%** when $\sigma = 9.28$ and between about **-0.05% and -1.20%** when $\sigma = 4.60$.
- With the exception of the United States in one specification, all countries are worse off under the tariff war relative to the factual tariff structure.

TABLE 5
WELFARE LOSSES FROM MOVING TO NASH TARIFFS

	$\sigma = 9.28$			$\sigma = 4.60$		
	Nash	Autarky	Relative	Nash	Autarky	Relative
ROW	−.28	−1.19	23.31	−.65	−2.23	29.22
European Union	−.03	−.41	6.42	−.18	−.86	21.20
Brazil	−.12	−.67	18.68	−.34	−1.13	30.18
China	−.36	−1.94	18.77	−1.20	−3.97	30.36
India	−.12	−.66	19.07	−.34	−1.08	31.60
Japan	−.08	−.39	20.87	−.30	−.87	34.86
United States	.01	−.35	−3.06	−.05	−.71	6.49

NOTE.—Entries under Nash are counterfactual welfare changes from moving to Nash tariffs, in percent; entries under Autarky are counterfactual welfare changes from moving to autarky, in percent; and entries under Relative are entries under Nash relative to entries under Autarky, in percent (computed from nonrounded values).

- These welfare losses are only a **fraction of autarky losses**: the Nash loss is roughly **6% to 35%** of the autarky loss depending on the country and elasticity assumption.
- So the paper's quantitative bottom line is balanced: noncooperative tariffs are meaningfully bad, but not remotely as catastrophic as shutting down trade altogether.

5.8 Further counterfactual: remaining gains from liberalization

Read:

- Even after major GATT/WTO tariff cuts, the model still predicts some room for further Pareto-improving liberalization.
- If countries choose tariff changes to maximize the mean welfare gain subject to no country losing and no tariff becoming negative, the average gain is about **0.11%** when $\sigma = 9.28$ and about **0.09%** when $\sigma = 4.60$.
- That is small in absolute magnitude, but it is positive and consistent with the idea that factual tariffs remain above the cooperative frontier.

6 Short summary

- **Conclusion.** Ossa provides a new-trade theory of trade agreements in which tariffs are used to attract manufacturing firms rather than to manipulate export prices.
- **Identification insight.** Reciprocity works by holding manufacturing trade balances constant, which prevents firm relocation. Nondiscrimination matters because it forces this logic to operate multilaterally.
- **Quantitative lesson.** Calibrated Nash tariffs are economically meaningful and historically plausible, but the welfare losses from a tariff war are still much smaller than the losses from autarky.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

- The paper establishes that GATT/WTO negotiations can be understood through a **production relocation externality** in a Krugman-style model with increasing returns.
- The central institutional result is that **reciprocity** internalizes this externality, while **nondiscrimination** makes that internalization work in multilateral settings.
- Quantitatively, the framework implies reasonable noncooperative tariffs and moderate gains from negotiated tariff reductions.

7.2 Economic intuition

- Governments care about tariffs because tariffs shift expenditure toward domestic manufacturing, which changes firm profitability, entry, and jobs.
- Left on their own, countries overuse tariffs because each one ignores the foreign harm created by pulling firms away from abroad.
- Reciprocity neutralizes the incentive to steal firms from others; nondiscrimination ensures that no third country is left outside the bargain.

7.3 Takeaway

This paper is best read as a clean conceptual alternative to the optimal-tariff view of trade agreements. Its big contribution is not merely to say that trade negotiations matter, but to say *why* they may matter in a world with increasing returns: countries negotiate because unilateral protection changes the geography of production. That interpretation fits the political rhetoric of trade disputes more naturally than pure terms-of-trade manipulation, and the calibration shows that it is also quantitatively plausible.